3.9 Laurent Polynomials

Definition 3.9.1 (Laurent polynomial ring)
Let A be a commutative ring. Define the ***laurent polynomial ring*** $A[X, X^{-1}]$ *to be the group ring $A[\mathbb{Z}]$. Denote an element as a formal sum*

$$f(X) \coloneqq \sum_{i=-\infty}^{\infty} a_i X^i$$

Proposition 3.9.2
Let B be an A-algebra and $b \in B^\star$ a unit. Then there is a unique A-algebra homomorphism $\mathrm{ev}_b : A[X, X^{-1}] \to B$ such that

$$\mathrm{ev}_b(X) = b$$

If the structural morphism $A \to B$ is injective, so is ev_b.

Proposition 3.9.3
Let A be a commutative ring. Then there is an isomorphism of rings

$$\begin{aligned} A[X, X^{-1}] &\xrightarrow{\sim} A[X]_X \\ \sum_{i=-r}^{\infty} a_i X^i &\to \frac{\sum_{i=0}^{\infty} a_{i-r} X^i}{X^r} \quad r = \min\{k \mid a_k \neq 0\} \vee 0 \\ \sum_{i=-r}^{\infty} a_{i+r} X^i &\leftarrow \frac{\sum_{i=0}^{\infty} a_i X^i}{X^r} \end{aligned}$$

Proof. Denote the maps by ψ and ϕ. Then ψ exists by (3.9.2) and ϕ exists by the universal property of localisation.■